Project scope

Week 1 | Farah Hammami | 13 November 2026

1. The research problem

My project is about an academic AI assistant for uterine fibroids. A language model can give a convincing answer even when the evidence is weak or missing. Retrieval-augmented generation (RAG) gives it documents to use, but finding a similar passage does not mean that the passage applies to the question. I want to study this gap, especially when different studies use different populations or outcomes.

2. Research question

With the same language model, source corpus and context budget, can evidence-aware re-ranking and claim verification reduce unsupported answers at comparable coverage? I will test whether better evidence selection helps before testing whether verification and abstention help. This is a controlled study, not a claim that I invented RAG or a clinical system.

3. Users and scope

The intended users are medical students and researchers looking for documented academic information. The assistant will focus on FIGO classification, diagnostic methods, symptoms, treatment options and fertility outcomes. Questions will mainly be in French; sources may be in English or French. Tunisian evidence will be included when it exists, with its actual context stated.

4. Three use cases

Use case	Example and expected behavior
Compare evidence	Compare myomectomy and embolization for quality of life and fertility. Show populations, outcomes, limits and source citations, without choosing treatment for a person.
Explain a concept	Explain FIGO types 2 and 3 and distinguish clinical pregnancy from live birth. Link each important claim to a source.
Handle missing evidence	If asked for a national Tunisian prevalence estimate without sufficient sources, state what is missing and give only supported information.

5. Three AI-proposed directions

Option	Why I keep or reject it
A. Standard document RAG	Feasible and useful, but a chatbot with citations alone does not isolate the reliability problem. I keep it as the M1 baseline.
B. Evidence-aware selective RAG	Selected direction. It lets me test applicability, claim support and refusal separately, with a manageable corpus and a controlled evaluation.
C. Knowledge graph / ontology QA	Interesting for FIGO and treatment relationships, but building and medically validating triples is too expensive for eight weeks. I leave it for future work.

6. What I deliberately exclude

Exclusion	Reason
Diagnosis, image analysis, personal prescriptions and triage	No patient data, clinical validation or safe basis for individual treatment decisions.
Patient-facing deployment	An academic prototype is not a substitute for a clinician.
Live web retrieval and restricted documents	A frozen and legally usable corpus makes the experiments traceable.
Fine-tuning, a graph and Arabic in the first version	More data, validation and computing time; French/English is enough for the initial experiment.

7. Risks and limits

The system may retrieve an outdated source, confuse outcomes, invent a citation, misjudge claim support or refuse too often. I plan to record source versions, use passage IDs, compare risk and coverage, and check outputs against source text. Without medical expert validation, I will measure document faithfulness rather than clinical accuracy. Only public sources and synthetic questions will be used.

8. My proposed contribution

The main value is not adding many components. It is designing a comparison where I can explain what each component changes. M0 is an LLM alone; M1 is standard hybrid retrieval; M2 adds evidence-aware re-ranking; M3 adds verification and a selective answer decision. FibroidQA and the ablation plan will make this question testable. No model results exist yet.